Optimization for Deep Learning (OPT4DL)

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Constrained Learning

Motivation



- We talked about numerous learning setups where

$$\min_{w} \mathbb{E}_z[\ell(w,z)] + \lambda h(w) \tag{1}$$

  where $h(w)$ was some regularization term and $\lambda \geq 0$ a regularization parameter
- How to determine λ? What if we want a certain condition is satisfied in a very strong sense?
- These questions lead to a general Constrained Learning setup

$$\min_{w} \mathbb{E}_z[\ell(w,z)] \qquad \text{s.t.} \quad h(w) \leq \tau \tag{2}$$

- The constraint $h(w)$ can itself be a stochastic function

$$h(w) = \mathbb{E}_{\tilde{z}}[\tilde{\ell}(w,\tilde{z})] \tag{3}$$

- We can set $\tau = 0$ without loss of generality by redefining $h(w) \leftarrow h(w) - \tau$

Motivation (Cont'd)



- Multi-class Neyman-Pearson classification: the loss of a prioritized class is minimized while the rest ones are below a certain threshold denoted by

$$\begin{aligned} \min_{w} \quad & f_1(w) \\ \text{s.t.} \quad & f_i(w) \leq \tau_i, \quad i = 2, \ldots, K \end{aligned} \tag{4}$$

where $f_i(w) = \mathbb{E}_{z_i}[\ell(w, z_i)]$ is the loss function of each class, and here class 1 is set as the prioritized one.

Motivation (Cont'd)



- Physics-Informed Neural Networks (PINNs): learning a DL model that approximates the solution to a partial differential equation (PDE).

$$\begin{aligned} \min_{w} \quad & \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, M_w(x_i)) \\ \text{s.t.} \quad & \mathbb{E}_{(x,t)\sim\mathcal{D}_{\textbf{physics}}}\left[\|\mathcal{N}[M_w](x,t)\|^2\right] = 0 \end{aligned} \tag{5}$$

where the distribution $\mathcal{D}_{\text{physics}}$ represents a set of points sampled from the domain where the physical constraints (e.g., differential equations) must be enforced and the operator $\mathcal{N}[M_w]$ represents the governing laws of the system, typically through a PDE.

Bringing the Constraint into the Objective



- Consider the problem

$$\min_{w} \mathbb{E}_z[\ell(w, z)] \qquad \text{s.t.} \quad h(w) \leq 0 \tag{6}$$

- If we define the indicator function of the constraint as

$$\mathbb{I}\{h(w)\} = \begin{cases} 0, & h(w) \leq 0 \\ \infty, & h(w) > 0 \end{cases} \tag{7}$$

we can equivalently solve

$$\min_{w} \mathbb{E}_z[\ell(w, z)] + \mathbb{I}\{h(w)\} \tag{8}$$

- The indicator function shows our amount of displeasure regarding the violation of constraint. Maximum displeasure leads to an equivalent problem, but not easily approachable. Other choices such as linear, quadratic, or the combination of the two lead to approaches
- This idea leads to penalty-based and Lagrangian-based formulations.

Weaker Displeasure Notations

- Penalty-Based method

$$\min_w \mathbb{E}_z[\ell(w,z)] + \frac{\rho}{2}([h(w)]_+)^2, \qquad \rho \geq 0 \tag{9}$$

where $[\cdot]_+ = \max(\cdot, 0)$

- Lagrangian (linear penalty) method

$$\min_w \mathbb{E}_z[\ell(w,z)] + \lambda h(w), \qquad \lambda \geq 0 \tag{10}$$

- Augmented Lagrangian method

$$\min_w \mathbb{E}_z[\ell(w,z)] + \lambda h(w) + \frac{\rho}{2}([h(w)]_+)^2 \quad \equiv \quad \min_w \mathbb{E}_z[\ell(w,z)] + \frac{\rho}{2}\left(\left[h(w) + \frac{\lambda}{\rho}\right]_+\right)^2 \tag{11}$$

Lagrangian



- The function $\mathcal{L}(w, \lambda) = \mathbb{E}_z[\ell(w, z)] + \lambda h(w)$ is called the Lagrangian associated with the original constrained learning task
- Note Lagrangian is linear in λ, thus, minimizing it w.r.t. w leads to a concave function which we call the Lagrange dual function

$$g(\lambda) = \min_w \mathcal{L}(w, \lambda) = \min_w \Big(\mathbb{E}_z[\ell(w, z)] + \lambda h(w) \Big) \tag{12}$$

Lagrangian (Cont'd)



- Lagrange dual function provides a lower bound on the optimal value of the original problem for each λ. For any feasible $\tilde{w}$

$$g(\lambda) = \min_{w} \mathcal{L}(w, \lambda) \leq \mathcal{L}(\tilde{w}, \lambda) = \mathbb{E}_z[\ell(\tilde{w}, z)] + \lambda h(\tilde{w}) \leq \mathbb{E}_z[\ell(\tilde{w}, z)] \tag{13}$$

  where we used $h(\tilde{w}) \leq 0$ for any feasible solution $\tilde{w}$ including the optimal one w^*.

- A natural question is: What is the best lower bound that can be obtained from the Lagrange dual function? This leads to the Lagrange dual problem

$$\max_{\lambda \geq 0} g(\lambda) \qquad \text{s.t.} \qquad \equiv \qquad \max_{\lambda \geq 0} \min_{w} \mathbb{E}_z[\ell(w, z)] + \lambda h(w) \tag{14}$$

- Under certain conditions (Slater's condition and strong duality), one can change the order of max and min and obtain an equivalent problem (to both dual and our original problem)

$$\max_{\lambda \geq 0} \min_{w} \mathbb{E}_z[\ell(w, z)] + \lambda h(w) \qquad \equiv \qquad \min_{w} \max_{\lambda \geq 0} \mathbb{E}_z[\ell(w, z)] + \lambda h(w) \tag{15}$$

Solving the Dual Problem via GD



- Let $\mathcal{L}(w, \lambda)$ denote the (augmented) Lagrangian function
- We can try to solve the min max (or max min) problem iteratively
- For $t = 1, \dots T$
  - $w_{t+1} = \min_w \mathcal{L}(w, \lambda_t)$
  - $\lambda_{t+1} = \max_{\lambda \geq 0} \mathcal{L}(w_{t+1}, \lambda)$
- The updates (after some approximations and under certain assumptions) can be interpreted as gradient descent on w and gradient ascent on λ:
  - $w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t, \lambda_t)$
  - $\lambda_{t+1} = [\lambda_t + \eta \nabla_\lambda \mathcal{L}(w_{t+1}, \lambda_t)]_+$

Switching Gradient Method (SGM)



- Consider our original problem

$$\min_{w} f(w) \qquad \text{s.t.} \quad h(w) \leq 0 \tag{16}$$

  and assume we are working with an iterative method.

- Note that the constraint need not be considered for an iterate w_t as long as $h(w_t) \leq \epsilon$. And we can just use GD or a similar method $w_{t+1} = w_t - \eta \nabla f(w_t)$
- On the other hand, if $h(w_t) > \epsilon$, we may want to prioritize the constraint over the objective by performing the following update as needed

$$w_{t+1} = w_t - \eta \nabla h(w_t) \tag{17}$$

- Effectively, depending on the amount of violation, we switch the gradient in GD
- For $t = 1, \ldots, T$
  - if $h(w_t) \leq \epsilon$ perform $w_{t+1} = w_t - \eta \nabla f(w_t)$
  - if $h(w_t) > \epsilon$ perform $w_{t+1} = w_t - \eta \nabla h(w_t)$

Switching Gradient Method (Cont'd)



- The method can be thought of as a kind of SGD for the dual problem.
- Consider $\min_w \max_{\lambda \geq 0} f(w) + \lambda h(w)$ and let $\lambda^* \geq 0$ be the maximizer. Then,

$$\min_w f(w) + \lambda^* h(w) \qquad \equiv \qquad \min_w \frac{1}{1+\lambda^*} f(w) + \frac{\lambda^*}{1+\lambda^*} h(w) \tag{18}$$

assuming λ^* is constant, for which the GD update is

$$w_{t+1} = w_t - \eta \left(\frac{1}{1+\lambda^*} \nabla f(w_t) + \frac{\lambda^*}{1+\lambda^*} \nabla h(w_t) \right) = w_t - \eta \mathbb{E}[\nabla \tilde{\ell}(w_t, z)] \tag{19}$$

where $z \in \{1, 2\}$, $\mathbb{P}(z = 1) = \frac{1}{1+\lambda^*}$, and $\mathbb{P}(z = 2) = \frac{\lambda^*}{1+\lambda^*}$, implying

$$\nabla \tilde{\ell}(w_t, z) = \begin{cases} \nabla f(w_t) & w.p. \quad \frac{1}{1+\lambda^*} \\ \nabla h(w_t) & w.p. \quad \frac{\lambda^*}{1+\lambda^*} \end{cases} \tag{20}$$